YUVA YODHA ENERGY TECH HACKATHON 2026 · CHALLENGE 04: SMART MANUFACTURING · PHASE 2

MeltWise

The working prototype

From the furnace meter to the operator's screen: a live energy copilot for induction-furnace foundries, with every result measured against ground truth.

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THE PROBLEM TODAY

The waste happens inside every melt, and nobody sees it

50–80

kWh per tonne swing between heats in the **same furnace**, from operating practice alone (BEE cluster study)

35–40%

of heat time runs at only 70–80% of rated power (BEE cluster study)

13–21%

above norms: induction-furnace energy per tonne in studied Indian foundries

About **5,000 foundry units** in India, 90% of them MSMEs, made **15.86 Mt** of castings in 2024–25. Most see energy only as a monthly bill, weeks after the melts that wasted it.

Five causes do the damage: **overshooting** the tap temperature, **holding** metal while moulds catch up, a **lid left open**, **under-powered** melting, and **sand and rust** in the charge.

WHAT WE BUILT

Six working parts, one pipeline

Edge gateway

Reads the furnace meter over Modbus TCP; buffers every reading in SQLite and forwards when the network is up.

Energy model

Physics terms recalibrated on the plant's own melts, plus gradient boosting on the charge mix. Explains every kWh by cause.

Live operator advisor

Bath-temperature soft sensor and six prompt types, in English, Hindi and Tamil, while the metal is still in the furnace.

Melt scheduler

Starts each heat so metal is ready when the moulds are, priced on the time-of-day tariff.

Manager dashboard

kWh per tonne by day and melt, avoidable kWh by cause, savings in rupees and CO_2.

Buyer carbon record

Scope 2 CO_2 for every batch from metered energy, CEA grid factor, CSV export.

Running code: Python 3.13 · NumPy · scikit-learn · FastAPI · SQLite · pymodbus · Chart.js. 14 automated tests. Source: the team's GitHub repository.

SOLUTION ARCHITECTURE

From the furnace meter to the operator’s screen

ON THE FURNACE

Signals

- Panel energy meter: kW, kWh (Modbus TCP)
- Lid limit switch (digital input)
- Operator app: charge, mix, dips, tap

IN THE FOUNDRY

Edge gateway

- Polls meter: one snapshot per read
- SQLite outbox: nothing lost offline
- Forwards in order when online

MELTWISE API (FASTAPI)

Models

- Energy model: physics + ML
- Soft sensor + live advisor
- Scheduler (ToD tariff, moulds)
- Carbon record (CEA factor)

FOR PEOPLE

Web app

- Operator: live prompts (en, hi, ta)
- Manager: kWh/t, causes, savings
- Planner: melt schedule
- Buyer: per-batch CO_2, CSV

Test harness (stands in for a plant until the pilot): a physics-based furnace simulator plays melts through a simulated Modbus meter at 40× speed. Re-running each melt with one practice gap removed gives the ground truth for every kWh of waste.

Schneider Electric fit: the gateway reads any Modbus meter, so EM6400NG or PowerLogic meters plug in by setting register addresses. Data can feed EcoStruxure or AVEVA.

HOW THE AI WORKS

Physics-informed machine learning, trained on the plant’s own melts

1 Measure practice from signals

Holding minutes, avoidable lid-open time, under-powered minutes and tap temperature, from meter power, the lid switch and the operator's log.

2 Recalibrate physics on this furnace

Each physics term (heat to spec, losses, overshoot) gets a coefficient learned from the plant's first 30 days, pulled toward the textbook value unless the data disagree.

3 Learn the charge

Gradient boosting on scrap mix, weight, grade and lining age gives the best-heat energy for a charge like this one. What's left over is charge quality: sand and rust.

4 Watch the bath live

A soft sensor tracks bath temperature from the energy balance and resets on every dip, so MeltWise can say “tap now” before the overshoot happens.

It learns the furnace

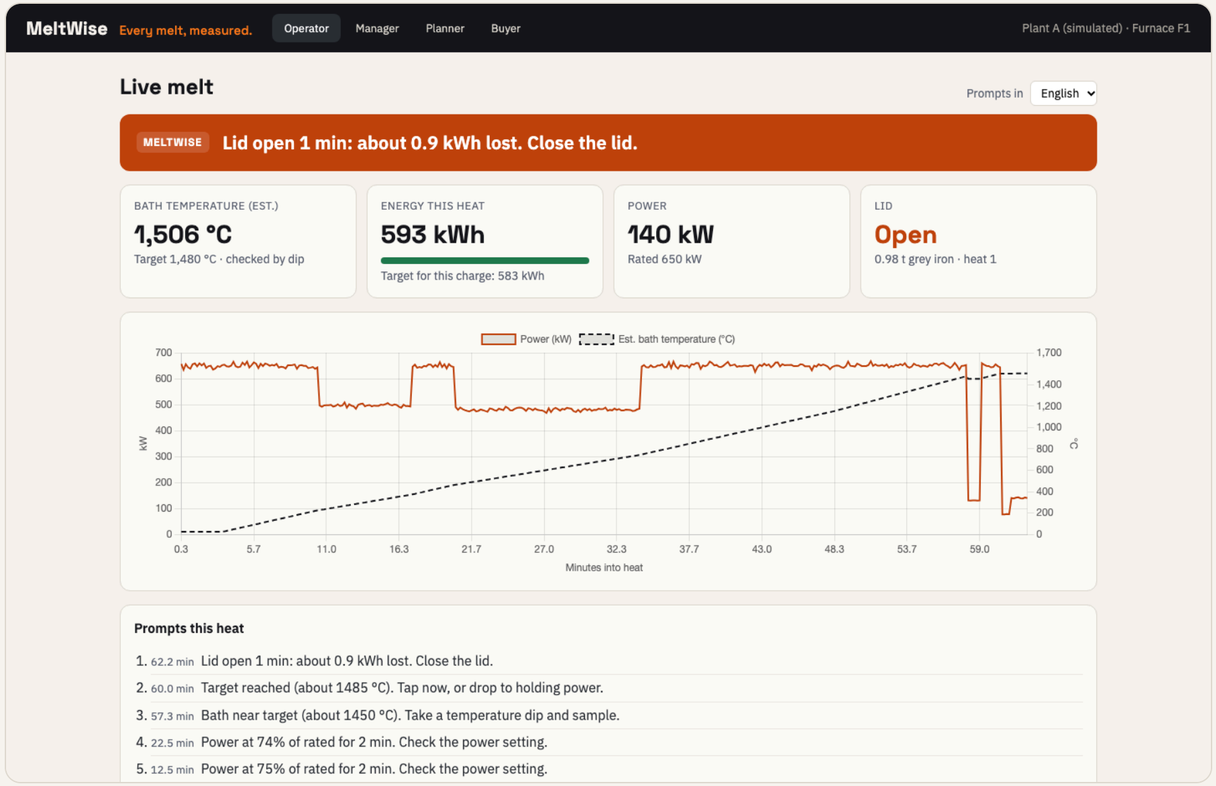
Plant A’s hidden constants (×textbook) vs what MeltWise learned from 240 melts

Constant	Learned	True
Losses while melting	1.15	1.12
Losses while holding	1.14	1.12
Open-lid losses	1.31	1.11
Coil efficiency (inverse)	1.00	1.00

Every kWh stays traceable to a named cause, so operators trust the prompt and owners trust the savings.

THE OPERATOR'S SCREEN

Prompts while the metal is still in the furnace



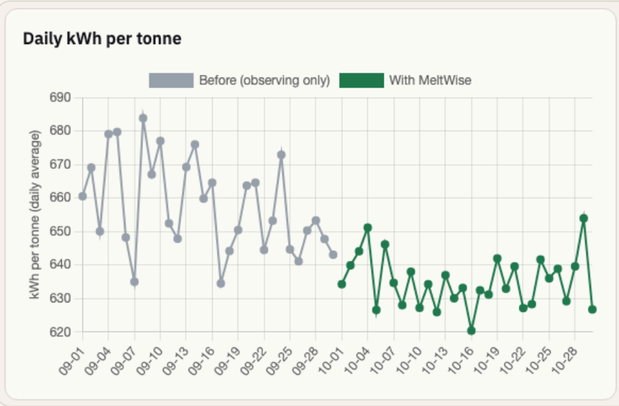


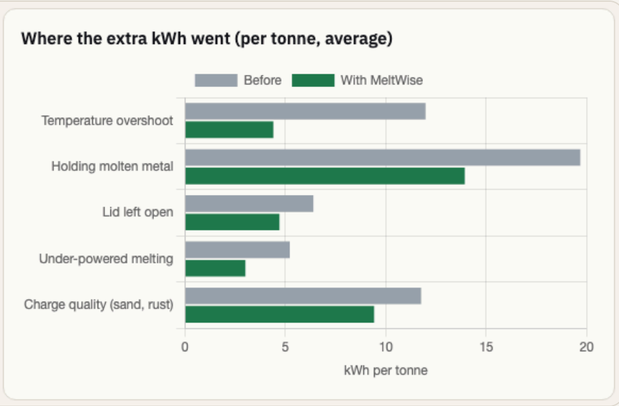
Live from the simulated meter, through the gateway. Six prompt types: under-power, dip now, **tap now**, lid open, holding, over budget, each with the kWh at stake. English, हिन्दी, தமிழ்.

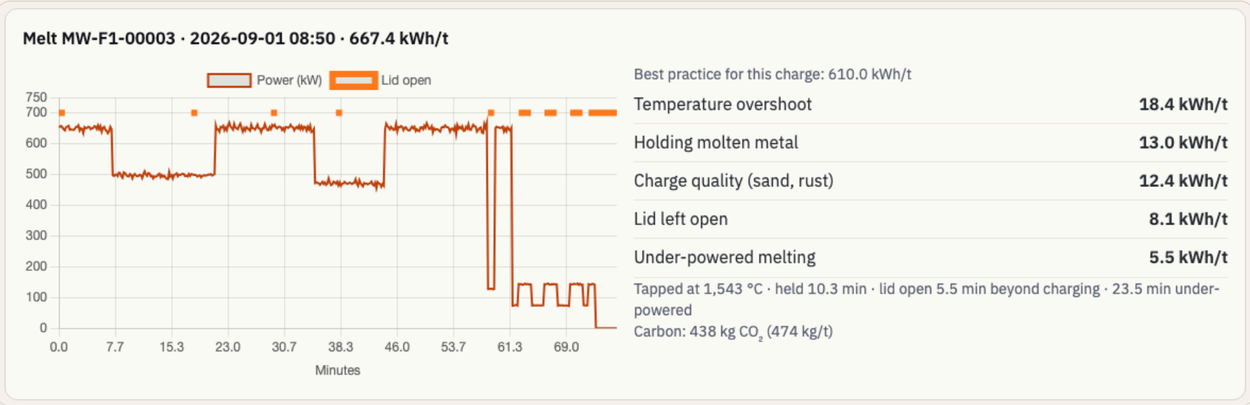
Hindi and Tamil text are first drafts, to be checked with operators in the pilot.

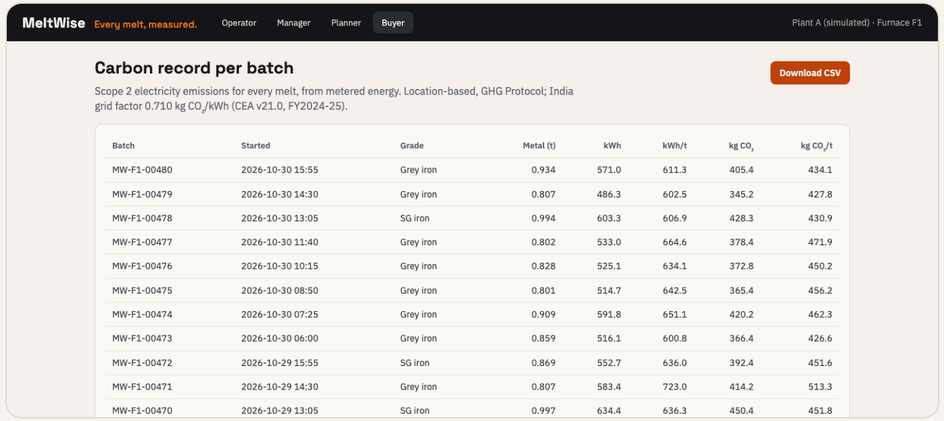
MANAGER, PLANNER AND BUYER

Where every kWh went, melt by melt









Manager: daily kWh/t, avoidable kWh by cause, savings in ₹ and CO_2. Any melt opens its power curve and breakdown: this one used 667 kWh/t against 610 for its charge.

Buyer: Scope 2 CO_2 per batch from metered energy (GHG Protocol, CEA 0.710 kg/kWh), as CSV. **Planner:** start times that cut holding.

HOW WE MEASURED

No plant yet, so we built one we can trust

1 · A simulator calibrated to published figures

	Simulator	Published
Typical melt today	648 kWh/t	650 kWh/t (our Phase 1 baseline, BEE cluster)
Best practice, same charges	588 kWh/t	593 kWh/t (US DOE best practice)
Avoidable loss per melt	65 kWh/t	50–80 kWh/t heat-to-heat swing (BEE)
Heat cycle	77 min	75 min (Phase 1 model plant)

2 · Ground truth for every kWh

Each melt is re-run with one practice gap removed; the difference is the true waste from that cause. MeltWise never sees these.

3 · Unseen plants

Every simulated plant has hidden loss and efficiency constants. MeltWise learns them from its first 30 days, then is scored on later melts.

4 · 14 automated tests

Including a network cut mid-melt: the gateway must deliver every reading, in order, once.

RESULTS: ACCURACY

MeltWise finds the waste, cause by cause

1.4%

error predicting each melt's kWh/t from its signals (300 unseen melts)

42 → 5 °C

tap temperature above spec when operators follow the “tap now” prompt

−49%

holding per heat with the scheduler (15.3 → 7.8 min)

0

readings lost when the network was cut mid-melt

Cause (kWh per tonne)	True mean	MeltWise	Correlation	Precision	Recall
Temperature overshoot	15.1	11.6	0.95	100%	98%
Holding molten metal	23.2	20.4	0.99	99%	95%
Lid left open	5.5	6.5	0.99	89%	97%
Under-powered melting	5.3	5.5	0.91	92%	98%
Sand and rust (not measured)	17.2	11.8	0.24	100%	75%

Plant A, 300 unseen melts; a cause counts as present above 3 kWh/t. Five more plants: practice-cause error 1.9–3.4 kWh/t. Sand and rust are only inferred, so they are under-counted.

RESULTS: SIMULATED PILOT

60 days at Plant A: 30 observing, 30 with MeltWise

Operators follow prompts	kWh/t before → after	Saved kWh/t	95% CI	Per year	CO_2 per year
40%	657.6 → 642.6	**14.9** (2.3%)	8.8–20.4	₹2.87 lakh	25.5 t
60%	657.6 → 635.0	**22.6** (3.4%)	17.1–27.7	₹4.33 lakh	38.5 t
80%	657.6 → 628.6	**29.0** (4.4%)	23.7–33.8	₹5.56 lakh	49.3 t

How it was measured: 240 melts before and 240 after, same plant and charge mix. Each prompt is followed only with the given probability, and each lever works through its mechanism: the operator taps on the soft sensor's call, closes the lid 30 s after the prompt, and so on.

Honest read: the conservative Phase 1 estimate (16.7 kWh/t) holds at 40–60% follow-through. Our Phase 1 “likely” 33.2 kWh/t needs about 90% (measured: 32.6 kWh/t). Sand and rust savings are still an assumption for a real pilot to test.

Per year: one 1 t furnace, 2,400 t a year, ₹8/kWh, 0.710 kg CO_2/kWh; before subscription.

COST AND BUSINESS MODEL

Cheap to build, fast to pay back

To build the prototype

₹0

cash spent: open-source stack, our laptops, simulated meter and plant

Next: a real pilot needs one kit and hosting (about ₹1,000 a month for the first plants).

To scale: per furnace

Item	Cost
Modbus energy meter (EM6400NG, class 0.5S)	₹17,289
CTs, wiring, lid switch, enclosure	₹15,000
Industrial edge gateway	₹20,000
Shop-floor display or tablet	₹15,000
Installation and training	₹20,000
Kit per furnace	**₹87,289**

Revenue and payback

Subscription **₹4,000 a furnace a month**, sold through foundry cluster associations; kit financed through SIDBI 4E or bundled with Schneider meters.

2.7 months

payback at the pilot's 60% result (₹3.85 lakh/yr net); 4.4 months at 40%

NEXT STEPS

Who uses it, and how it rolls out

NOW – JAN 2027

First real furnace

- One foundry via a cluster association
- Install kit; 30 days observing
- 30 days with prompts: measure kWh/t
- Check translations with operators

FEB – JUN 2027

10 furnaces, one cluster

- Retrain the model per furnace
- Supplier charge-quality reports
- WhatsApp summaries for owners
- Measure follow-through, refine prompts

LATE 2027 →

Scale

- Cluster associations as channel
- SIDBI 4E and BEE schemes for kits
- Schneider meters, EcoStruxure link
- Steel and aluminium melt shops

Users: the furnace operator (prompts on a shop-floor screen), the shift supervisor and plant manager (dashboard and schedule), the owner (monthly savings), and buyers (carbon record per batch).

BROADER IMPACT

If 1,000 foundries reach the pilot’s 60% result

54 GWh

electricity saved a year

₹43 crore

a year back to MSME foundries

38,455 t

of CO_2 avoided a year (grid factor 0.710)

Economic: energy is 15–30% of cost in energy-intensive SMEs; a saving that pays back in months protects thin margins and jobs.

Environmental: per-batch, metered CO_2 records help small suppliers answer buyers’ carbon-disclosure requests.

Social: prompts in the operator’s own language put the know-how of the best shifts in every operator’s hands.

MeltWise · Every melt, measured.

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